

## SUMMARY

Results-driven Electronics and Communication Engineering student with a solid foundation in building IoT systems, embedded solutions, and real-time monitoring applications. Skilled in leveraging hardware-software integration expertise and technical proficiency across Arduino, ESP32, C/C++, and Python to deliver high-quality, innovative projects. Completed 4 internships spanning Embedded Systems, IoT, VLSI Design, and Web Development, with hands-on project experience. Committed to driving real-world impact, contributing to cutting-edge embedded technology, and expanding professional expertise through collaborative opportunities in the electronics and tech industry.

## TECHNICAL SKILLS

**Core Domains** : Internet of Things(IoT), Embedded System, PCB Design, VLSI Design

**Languages** : C, C++, Html, Python

**Hardware** : Arduino, Esp32, Sensors, I2C

**Tools** : Arduino IDE, Proteus, VS Code, GitHub, AdobePhotoshop

**Soft Skills** : Communication & Teamwork, Time Management, Adaptability & Flexibility, Leadership Skills

## EDUCATION

2023 - 2027 **B.E. – Electronics and Communication Engineering**  
Gnanamani College of Technology | CGPA: 8.0 / 10

2022 - 2023 **H.S. – Maths, Computer Science**  
TN State Board | Scored 75.5%

## PROJECTS

### Smart Home Automation System (IoT)

Stack: ESP32, Relay, Wi-Fi Module, Sensors

- Designed and developed an IoT-based smart home system using ESP32 and sensors. Enabled remote monitoring and control of home appliances.
- Implemented real-time automation logic to improve efficiency.

### AI & IoT Based Healthcare Monitoring System

Stack: Esp32, IoT Sensors, Python

- Built an IoT-enabled health care monitoring system integrating sensors with AI-based data analysis. Monitored vital parameters such as pulse and temperature with real-time alerts.
- Enhanced health data tracking and intelligent decision support.

### Distance Measurement Using Ultrasonic Sensor

Stack: ArduinoUno, HC-SR04 Sensor, LCD Display

- Developed a distance measurement system using ultrasonic sensor, Arduino, and LCD display.
- Built a real-time distance measurement system with  $\pm 1$  cm accuracy over a 2–400 cm range. Displayed live readings on 16x2 LCD and calibrated sensor output for reliable environmental use.

### AI-Based Soil Data Communication and Disaster Management

Stack: ESP32, IoT, Embedded C, Sensors, Rule-Based AI

- Developed an IoT-based soil monitoring system using ESP32 and sensors to collect real-time environmental data for agriculture and disaster prevention.
- Implemented rule-based AI for intelligent analysis of soil, rain, and water levels to detect risks like floods and landslides and send alerts.

## INTERNSHIP EXPERIENCES

Dec 2025 – Jan 2026

**Embedded Systems Intern**  
UnitedSoftTech | Salem

- Worked on embedded system design and development, gaining hands-on experience in microcontrollers, sensors, and circuit interfacing.
- Implemented basic embedded programming concepts for real-time applications.

Oct 2025 – Dec 2025

**Very Large Scale Integration - intern**  
Smarted Innovation | Bangalore | Online

- Gained practical exposure to VLSI fundamentals including digital logic and CMOS basics.
- Understood IC design flow such as specification, design, and basic verification.
- Strengthened core knowledge relevant to semiconductor and VLSI industries.

Jul 2025

**Internet of Things - intern**  
Emglitz Technologies | Salem

- Learned and applied IoT fundamentals using Arduino and ESP32. Integrated sensors for real-time monitoring and control applications.
- Developed basic IoT mini-projects for practical implementation.

Jan 2025

**Web Development - intern**  
Odugaa Tech | Salem

- Gained practical exposure to VLSI fundamentals including digital logic and CMOS basics.
- Understood IC design flow such as specification, design, and basic verification.
- Strengthened core knowledge relevant to semiconductor and VLSI industries.

## CERTIFICATIONS

**PCB Design Workshop** : Mastering PCB Design: A Hands-on Training Workshop” by IndiGuard Systems Pvt. Ltd. at Sep 2024 Gnanamani College of Technology.

**PROCOMPANION Workshop** : Attended a technical workshop at National Institute of Technology (NIT) on Procompanion, gaining Sep 2024 | Sponsor Qualcomm exposure to advanced technical concepts and hands-on learning.

**Workshop** : Attended National Level Symposium on "Generative AI for Electronics and Communication" at KSR Sep 2024 Engineering College, gaining insights into AI-driven innovations in the ECE domain.

**Project Expo** : Participated in i-SYM'26 (IEEE) Project Expo at Hindusthan College of Engineering and Technology, Mar 2026 showcasing a technical project and enhancing presentation skills.

Completed online and hands-on learning programs related to technology and skill development. presented technical ideas through PPT presentations and actively engaged in inter-college workshops to improve communication, teamwork, and technical knowledge.

## Achievements & Activities

Jan 2025

**Paper Presentation** : Won 2nd Prize in Paper Presentation on Haptic Technology at AVS Engineering College, Jan 2025 Technical Symposium | 2<sup>nd</sup> prize demonstrating strong research and technical communication skills.

Sep 2025

**SIH** : Participated in Smart India Hackathon (SIH) with an innovative IoT-based solution, collaborating in a Sep 2025 Government Hackathon team to address real-world problems.

## Languages

English  
Tamil